CNENGR202 Instrumentation and Control

Part 3 L9 – Final Lab

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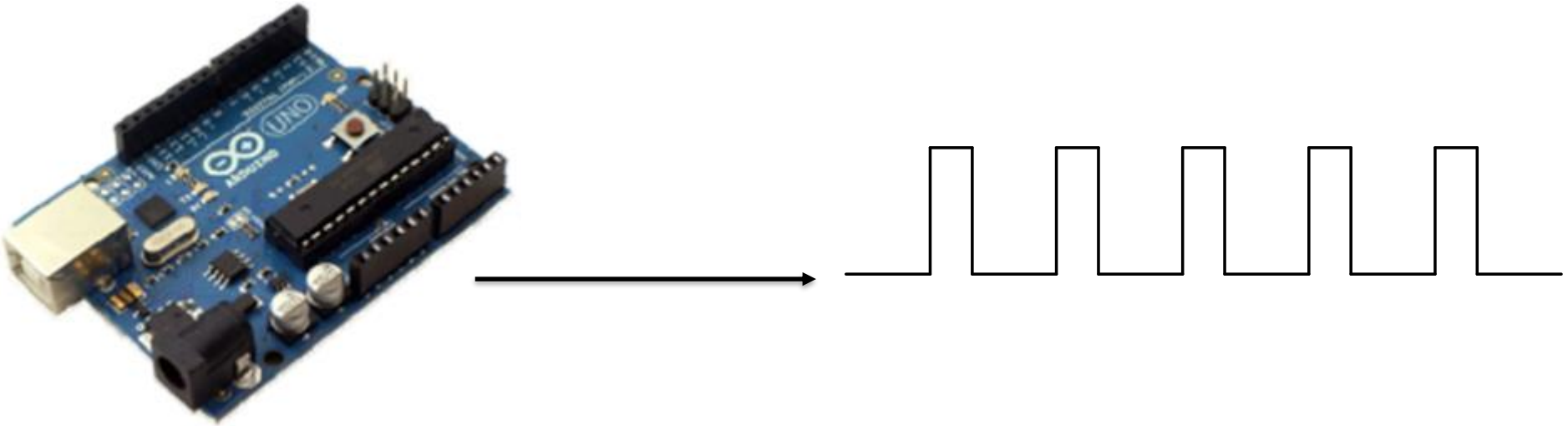
The task

- I want you to create a PWM waveform to drive motor



The task

I want you to generate PWM waveform



Trim pot to Arduino

- Connect trim pot to Arduino
  - Use trim pot as potential divider
  - Connect to Analogue to Digital Converter





Motor to Arduino

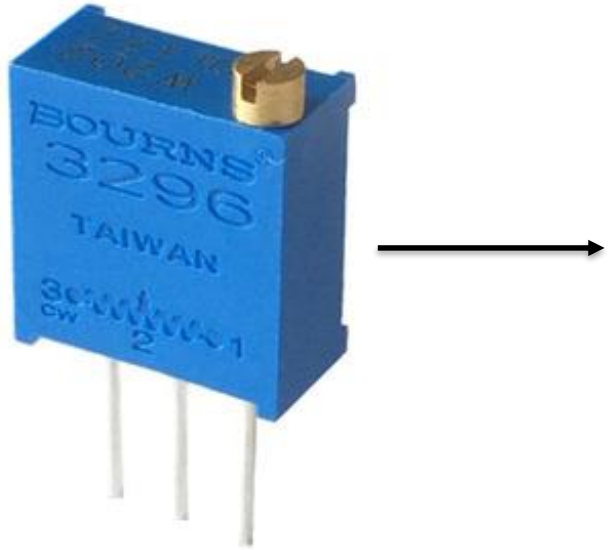
- Connect oscilloscope to Arduino
- Observe PWM output

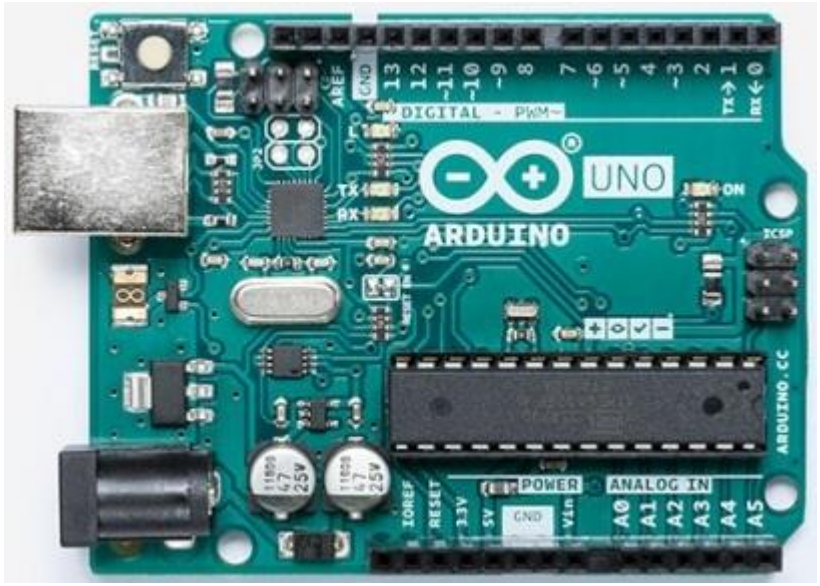


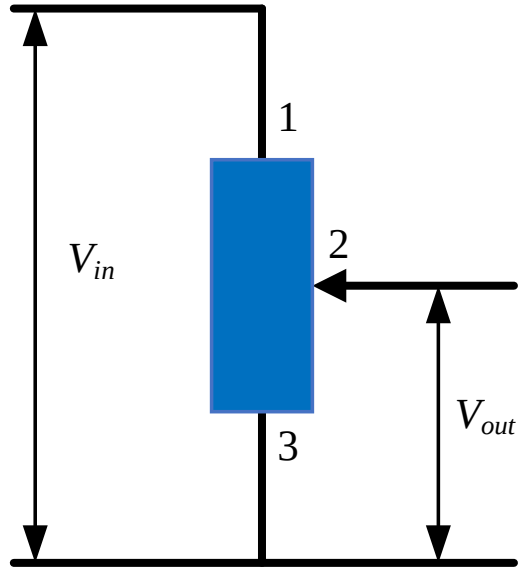


Trim pot to Arduino

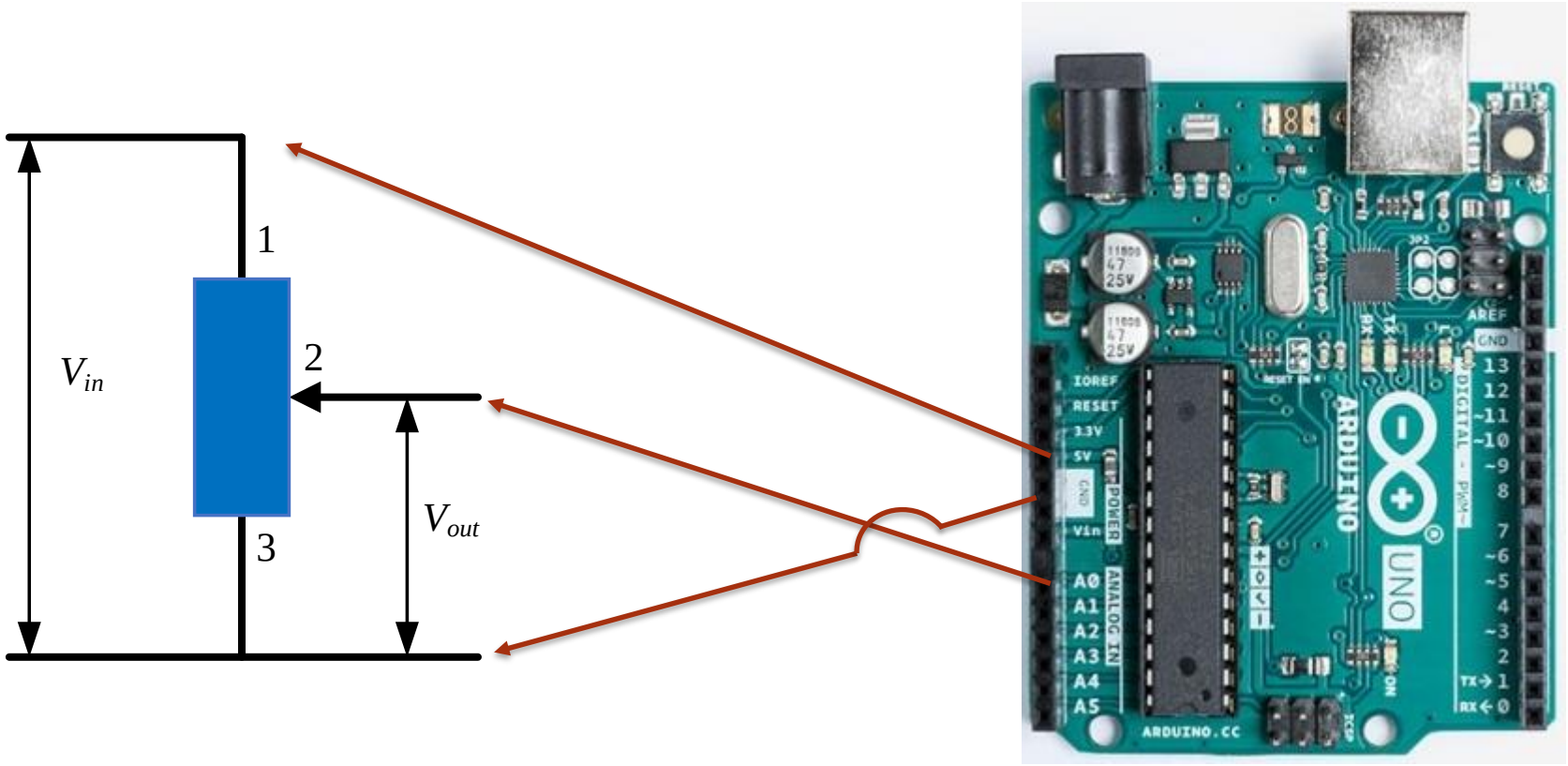
- A trim pot is variable potential divider
- Measure this voltage with the ADC





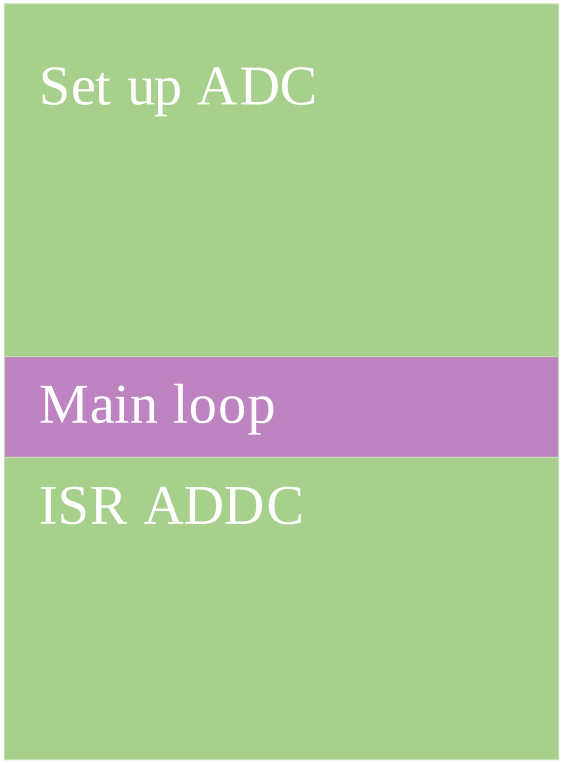


Connections



Read ADC

- Three stages to the program
  1. Setup the ADC using it's control registers to interrupt after conversion
  2. A main loop to keep the program running
  3. ADC conversion complete ISR



Set up the ADC

1. Enable ADC
2. Start the ADC
3. Set it to free running mode
4. Select frequency
5. Enable conversion complete interrupt

ADC Status and Control register

	7	6	5	4	3	2	1	0
ADCSRA	ADEN	ADSC	ADATE	ADIF	ADIE	ADPS2	ADPS1	ADPS0
Initial Value	0	0	0	0	0	0	0	0

Ext I/O Registers

- ADEN: Enables the ADC
- ADSC: Starts a conversion
- ADATE: Auto trigger enable
- ADIF: Interrupt Flag
- ADIE: Interrupt enable
- ADPSn: Prescaler Select Bits

Prescaler

- To give ADC time to achieve 10bit accuracy need clock between 10kHz and 200kHz
- 16MHz/128 = 125kHz

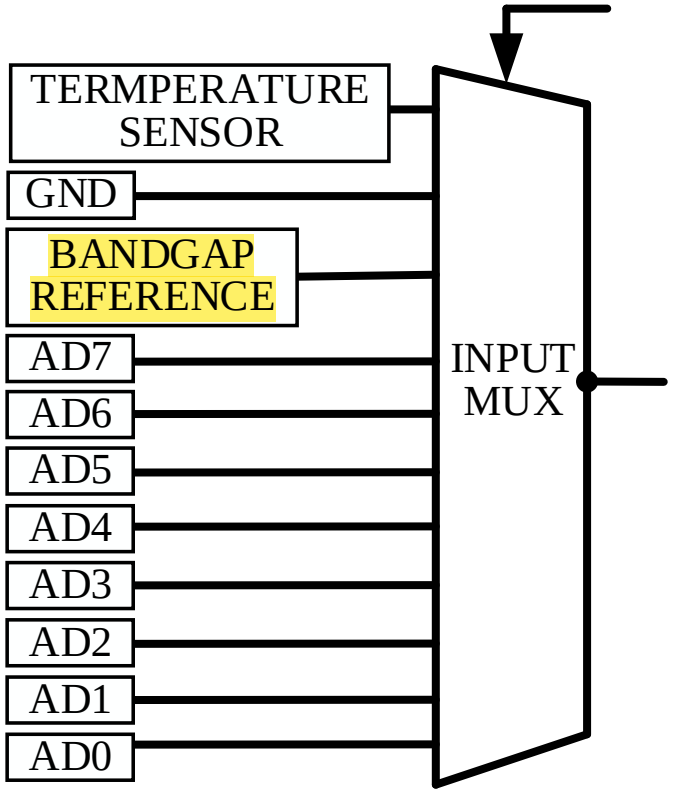
ADC Prescaler Selections

ADPS2	ADPS1	ADPS0	Division Factor
0	0	0	2
0	0	1	2
0	1	0	4
0	1	1	8
1	0	0	16
1	0	1	32
1	1	0	64
1	1	1	128

ADC Input Channel Select

	7	6	5	4	3	2	1	0
ADMUX	REFS1	REFS0	ADLAR	-	MUX3	MUX2	MUX1	MUX0

Ext I/O Registers



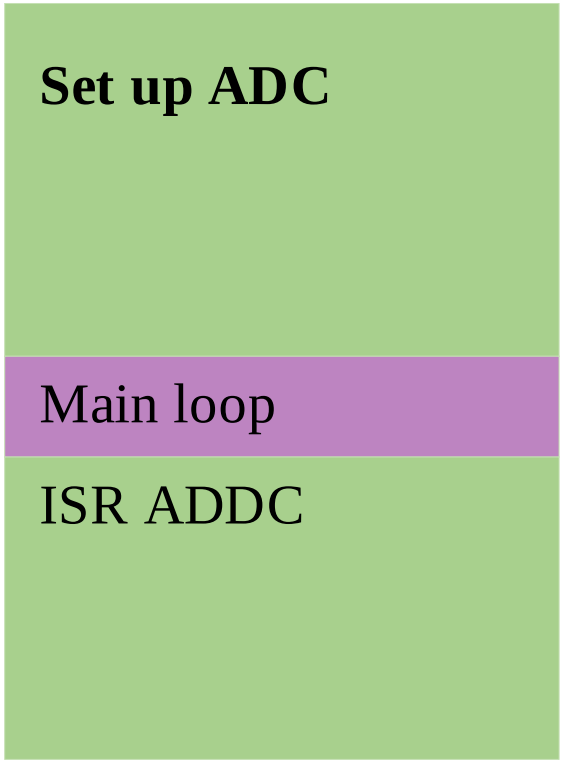
1 For temperature sensor

Input Channel Selections

MUX3-0	Single Ended Input
0000	ADC0
0001	ADC1
0010	ADC2
0011	ADC3
0100	ADC4
0101	ADC5
0110	ADC6
1110	1.1V(V_{BG})
1111	0V(GND)

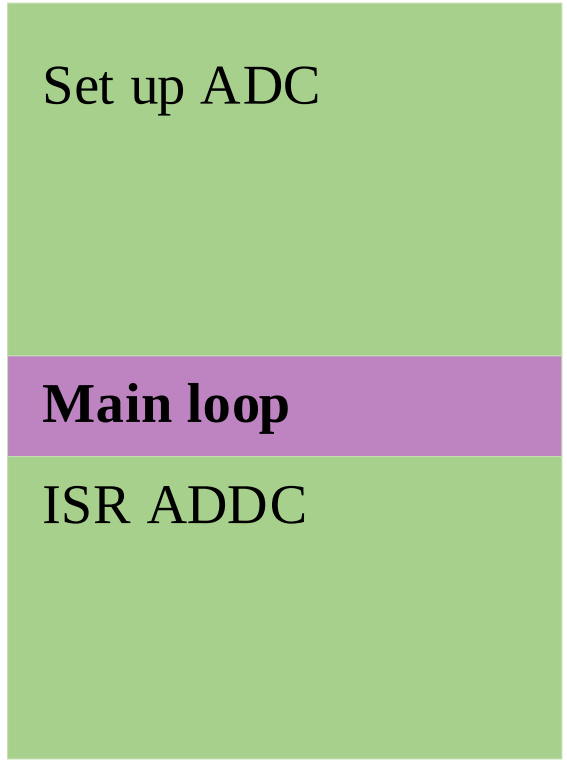
Write the Program

- Three stages to the program
  1. **Setup the ADC using it's control registers**
  2. A main loop to keep the program running
  3. ADC completion interrupt



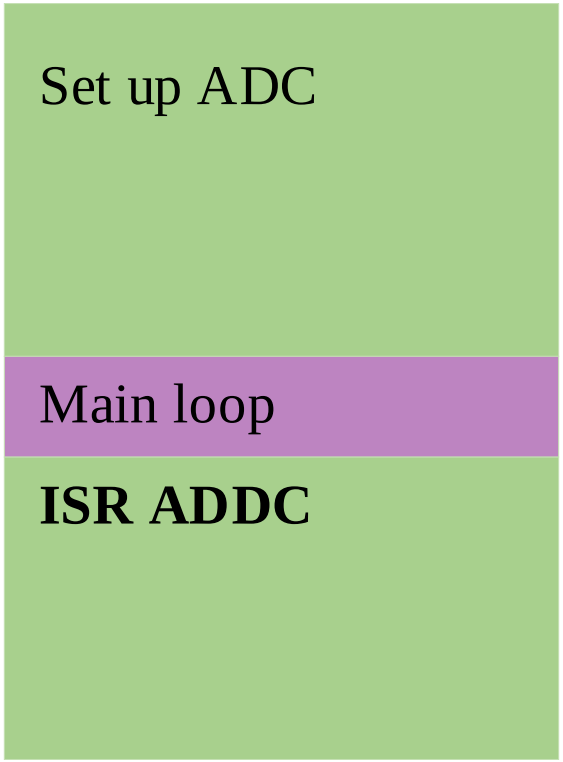
Write the Program

- Three stages to the program
  1. Setup the ADC using it's control registers
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Write the Program

- Three stages to the program
  1. Setup the ADC using it's control registers
  2. A main loop to keep the program running
  3. **ADC completion interrupt**

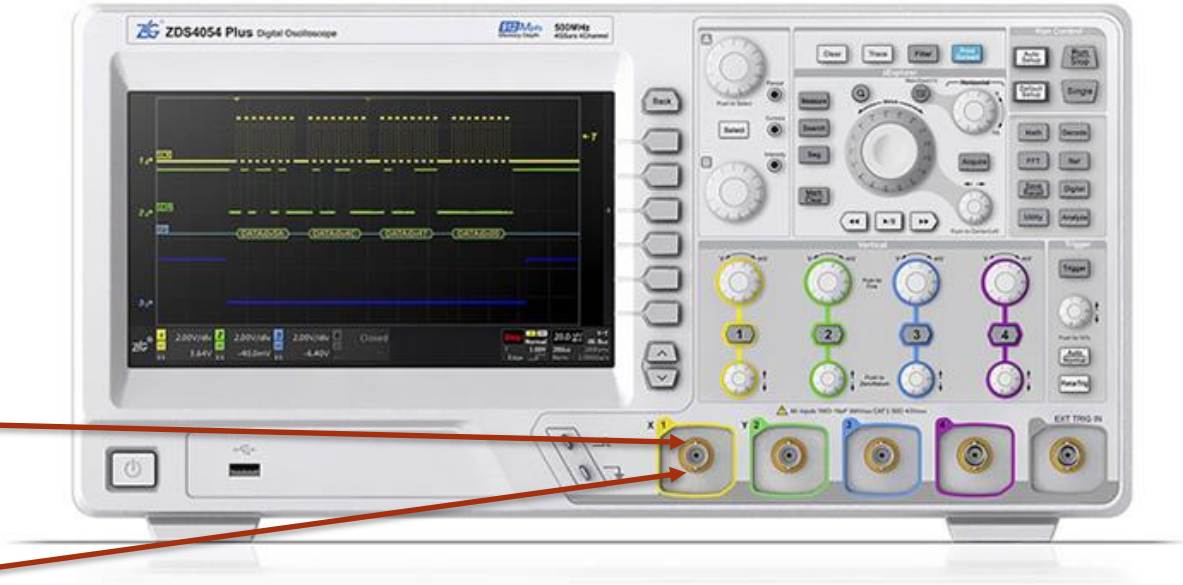


Connect your oscilloscope



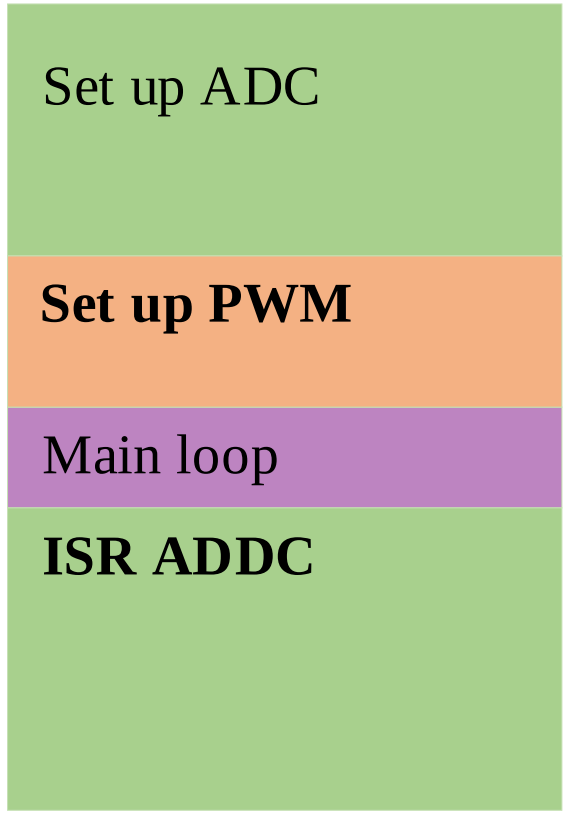
Ground

Signal



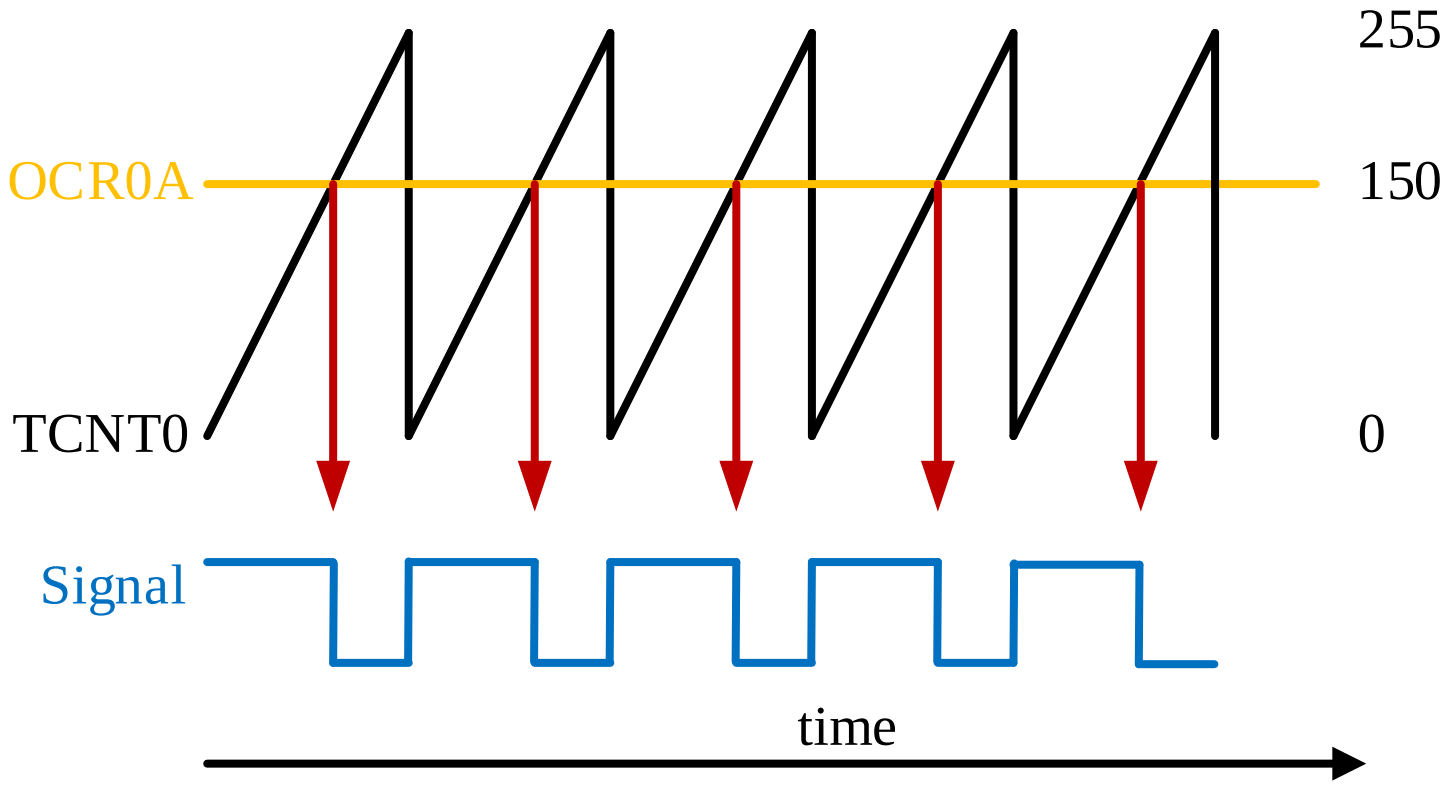
Setup PWM

- Four stages to the program
  - Setup the ADC using it's control registers
  - **Setup the PWM**
  - A main loop to keep the program running
  - ADC completion interrupt **to adjust PWM**



Fast PWM

- Counts up to 0xFF or another maximum value.
- When counter passes OCR0A output goes low.
- Counter overflows output goes high and starts again.

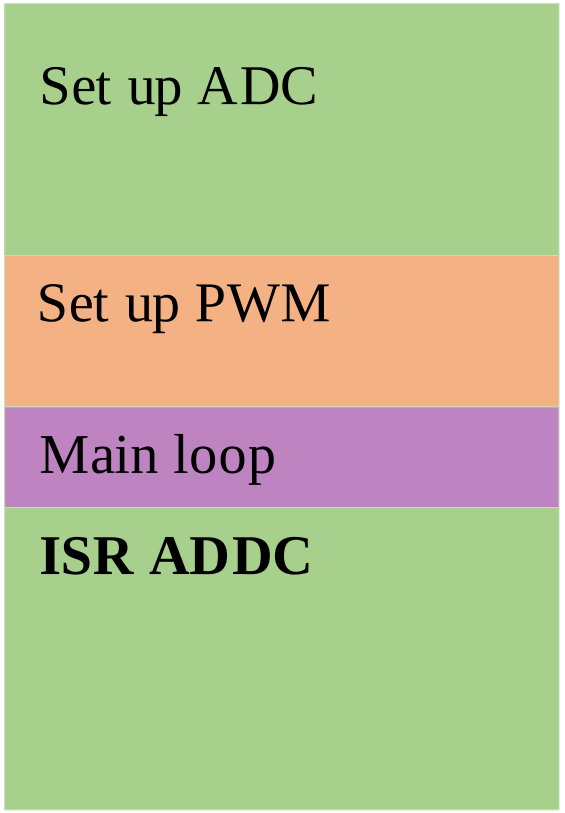


Fast PWM

- Implementation in PWM lecture
- Configure TCCR0A and TCCR0B

Write the Program

- Four stages to the program
  - Setup the ADC using it's control registers
  - Setup the PWM
  - A main loop to keep the program running
  - **ADC completion interrupt to adjust PWM**



Change PWM Duty Cycle

- Copy the value from ADCH/ADCL to OCR0A and so control the speed of the motor with the trim pot.

Assessment - I

I would like you to prepare a report covering what you have achieved in the four laboratory sessions taken in weeks 13 - 16. Think of this in the context of condensing your knowledge, achievements and results into the report constraint. In particular, the report should be:

- Technically accurate and consistent with the documentation that accompanies the Atmel 328P microcontrollers used in the labs
- Attractive to the reader and accessible to all with an interest in the use of these devices
- You should consider including
  - A very short piece on how and where microcontrollers are used
  - Brief details of the Atmel sufficient to set the context of the labs including its capa bilities
  - An example programme showing and highlighting the important elements and a description of the development cycle.
  - Consider including an example simple circuit diagram

Assessment - II

- Of a professional Engineering standard.
- **A complete record of what you have achieved in the labs**

The mark scheme is as follows:

	Marks
The extent of the study: how much have you achieved?	10
Appearance of the report: is it of a professional standard, accurate and complete?	10
Does it meet the stated requirements?	10
Is there evidence of additional thinking around the laboratory study?	10

To be hand in 9th January 2022 – Please submit on Moodle as a PDF document

Example Layouts

